

From Robots to Creatures: The Influence of Pedagogical Agent Design on Student Motivation and Learning in IVR Education

Linjing Sun*

Boon Giin Lee†

Matthew Pike‡

Sen Yang§

David Chieng||

Faculty of Science and Engineering
University of Nottingham Ningbo China

ABSTRACT

Pedagogical agents (PAs) have been extensively employed in immersive virtual reality (IVR) educational settings, yet their specific attributes are not thoroughly investigated. This study explores the impact of PA appearance and knowledge delivery methods on student learning in IVR environments. The findings reveal that the flying robot PA is regarded as the most credible, while cute creature PAs increase motivation but cause some distraction. The integrated text and audio multi-modality approach resulted in the best knowledge retention, though some participants experienced a slower learning pace. These insights are significant for designing and using PAs in IVR educational settings.

Index Terms: Immersive virtual reality, pedagogical agent, education.

1 INTRODUCTION

In recent decades, pedagogical agents (PAs) have become increasingly integrated into educational settings. PAs are virtual agents adapted to digital learning environments to aid in the educational process. Studies suggest that PAs improve interactivity and promote social engagement in learning environments [2]. By taking on roles such as peers, tutors, or coaches, and offering the necessary feedback and prompts, PAs help to promote deeper learning engagement and comprehension, thus improving student motivation. The introduction of immersive virtual reality (IVR) technology expands the possibilities of PAs by offering immersive experiences and high-quality interactive content. Studies have shown that PAs in IVR settings provide more personalized learning, which reduces student anxiety and ensures behavioral consistency [4]. However, most IVR studies on PAs focus on the integration of PAs with various game components, rather than investigating the impact of specific characteristics of individual PAs on educational outcomes.

The visual and aesthetic design of PAs is significant for their effectiveness, as studies indicate that the appearance of PAs influences student engagement, perception, and performance [5]. Humanoid PAs are most commonly adopted in the IVR learning environment due to the approachable nature of natural faces [6], although such forms can trigger the “uncanny valley” effect [3], which causes the learner discomfort. Consequently, there is interest in non-humanoid PAs, like creature-like avatars, which can enhance perceived pleasure, social interaction, and influence while reducing technical constraints.

The modality used by PAs to convey educational content significantly influences the learning experience. Multiple studies have

shown that text and audio are the predominant modalities utilized by PAs. Studies suggest that text significantly increases memory retention, self-efficacy, and focus compared to audio [1]. In contrast, audio is better at promoting engagement, emotional connection, and immersion. However, integrating text and audio to deliver knowledge in IVR settings has not been fully explored.

This work sought to explore different PA designs and knowledge delivery methods to improve student motivation and learning outcomes within IVR contexts. By analyzing various PA appearances (flying robots, humanoid and non-humanoid forms) and evaluating the delivery modes of knowledge (text, audio, and text+audio), this study offers insight into their impact on student engagement and learning retention in IVR environments.

2 METHODOLOGY

This study implemented both between-subjects and within-subjects experiments, which are carried out simultaneously. In the between-subjects experiment, subjects were randomly divided into three groups, each interacting with one of the three types of PA avatars in IVR learning settings: humanoid (Nany), flying robot (Cheese), or creature (Minccino), as shown in Fig. 1. Meanwhile, in the within-subjects experiment, each subject participated in three consecutive educational sessions on water filtration, water electrolysis, and hydrogen-oxygen fuel cells using different content delivery methods: text, audio, and combination of text and audio (text+audio), as presented in Fig. 2. Following each session, the subjects participated in game exercises to apply their knowledge practically.

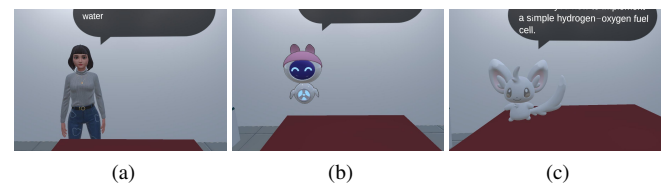


Figure 1: Design of the between-subjects study featuring three PA appearances in IVR learning settings: (a) humanoid (Nany), (b) flying robot (Cheese), and (c) creature (Minccino).

Thirty subjects were randomly selected from the campus student population through volunteer sampling. Subjects received information about the study and were requested to sign the consent form, with the agreement that they were allowed to withdraw at any time and that their data would not be recorded. The subjects included 19 female students (63.3%) and 11 male students (36.7%), aged 18 to 27 years with a mean of 22.07 years. More than half of the subjects reported having IVR experiences. The study complied with the university research ethics guidelines and was approved by the faculty ethics committee.

Upon completion of the experiments, subjects were required to complete several questionnaires, among them, the self-designed Knowledge Retention Questionnaire (KRQ) to evaluate learning

*e-mail: linjing.sun@nottingham.edu.cn

†e-mail: boon-giin.lee@nottingham.edu.cn

‡e-mail: matthew.pike@nottingham.edu.cn

§e-mail: sen.yang@nottingham.edu.cn

||e-mail: david.chieng@nottingham.edu.cn

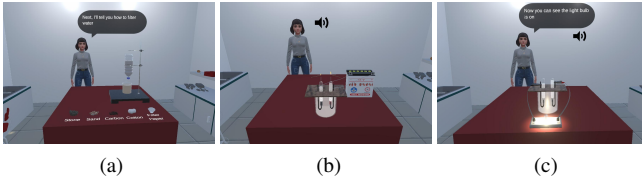


Figure 2: Design of the within-subjects study featuring three knowledge delivery methods in the IVR learning environments for three different sessions sequentially: (a) text (water filtration), (b) audio (water electrolysis), and (c) text+audio (hydrogen-oxygen fuel cells).

performance, the Student Motivation Toward Science Learning (SMTSL) Questionnaire to evaluate learning motivation, and the User Experience Questionnaire (UEQ+) to evaluate learning experience in IVR learning environments. Generally, these questionnaires were used to assess general knowledge retention in subjects through different types of learning sessions. Subsequently, they participated in an in-depth interview to offer further insight into their learning experiences with various combinations of PA appearances and methods of delivering knowledge.

The study was approved by the Research Ethics Committee of the Faculty of Science and Engineering of the University of Nottingham Ningbo China.

3 RESULTS AND DISCUSSION

3.1 Analysis of Learning Impact with Different PAs

Table 1 summarizes the scores of the knowledge retention questionnaire for three different PAs. The KRQ results showed a normal distribution of mean ratings, supported by the Shapiro-Wilk test ($p > 0.05$). ANOVA (Welch test) indicated significant differences between groups ($p < 0.01$). The Games-Howell test did not highlight significant differences between Nany and Cheese ($p = 0.016$), but Minccino differed significantly from both Nany ($p < 0.001$) and Cheese ($p < 0.001$), indicating a significant influence of Minccino on knowledge retention.

Table 1: Summary of mean ratings for knowledge retention across three PA appearances for different questionnaires: KRQ (maximum score of 35), SMTSL (maximum score of 42), and UEQ+ (range -3 to 3).

PA	KRQ μ (σ)	SMTSL μ (σ)	UEQ+ μ (σ)
Nany	27.97 (2.82)	34.20 (0.81)	4.52 (3.41)
Cheese	26.10 (2.71)	32.19 (0.61)	5.17 (2.32)
Minccino	31.00 (2.54)	34.90 (0.76)	4.13 (3.61)

The SMTSL analysis showed significant differences between Nany and Cheese ($p < 0.032$) on the scale of active learning strategies, as well as between Minccino and Nany ($p = 0.028$) on the scale of learning environment stimulation. Minccino consistently outperformed the other PAs, and the subjects stated that its depiction as a cute creature promoted a positive learning attitude and increased motivation. The UEQ+ results showed that Cheese ranked highest for trustworthiness of content and quality of content dimensions, with most subjects perceiving its appearance as an indicator of high intelligence through interview feedback. In contrast, the appealing look of Minccino attracted student attention, but did not increase their trustworthiness.

In general, the results suggest that adapting the presentation of PAs to align with student preferences can greatly improve both the engagement in learning activities and the motivation within IVR settings.

3.2 Evaluation of Knowledge Delivery Methods

The KRQ results indicated a significant impact of the knowledge delivery method on learning outcomes, with the text+audio method ($\mu = 32.00$, $\sigma = 2.54$) outperforming the text ($\mu = 28.00$, $\sigma = 2.82$) and the audio ($\mu = 24.50$, $\sigma = 2.72$) methods. Likewise, the findings of SMTSL and UEQ+ indicated that text+audio was the preferred learning method, particularly improving learning efficiency and achievement of objectives.

During interviews, subjects expressed a higher degree of confidence in their learning when using the text+audio method, implying an improved ability to retain and engage with the knowledge presented. Although some subjects with a high reading speed preferred the text-only method, many subjects stated that combining text and audio improved both concentration and material retention.

These findings highlight that using multi-modality inputs can improve learning effectiveness and outcomes, particularly when aligned with individual PA preferences.

4 CONCLUSION

This study explores the impact of different PAs and knowledge delivery methods on knowledge retention, learning motivation, and the overall learning experience within IVR environments. The findings observed that the students showed a preference for PA with a cute appearance, highlighting the importance of aesthetic appeal in increasing engagement levels and learning outcomes. On the other hand, the text+audio delivery method yielded the highest level of knowledge retention, providing indication of the advantages associated with using multi-modal inputs in the educational development.

This study had certain limitations, such as participants experiencing fatigue and cybersickness when participating in IVR learning environments using a VR headset. Furthermore, the research did not thoroughly examine the use of randomized delivery methods as a means to deliver knowledge. Future work will aim to broaden the scope of the game and increase the sample size by involving subjects from a wider range of educational backgrounds. In addition, integrating artificial intelligence into PA as smart agents is a promising approach to improve immersive interactions with learners. However, more studies are needed to develop intelligent dialogue and adaptable teaching materials to vary student cognitive abilities.

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